COMPUTER SCIENCE

O Level Notes

Syllabus 2210 / 0478

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O Level Computer Science

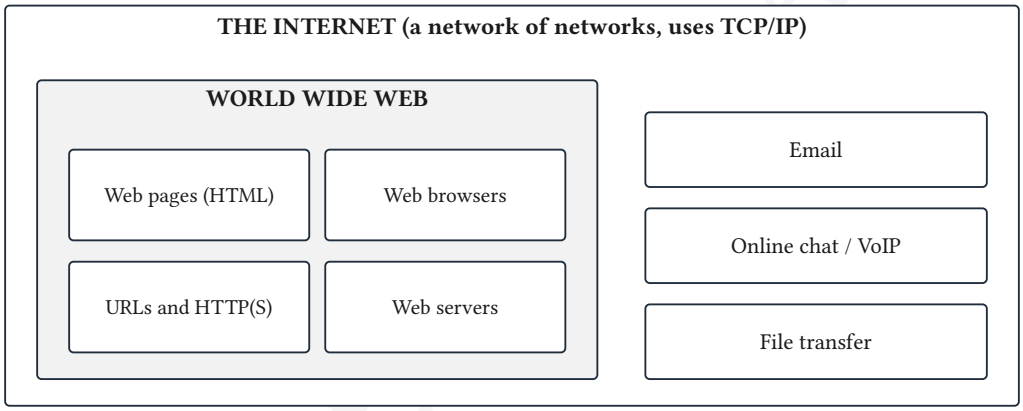
Chapter 5: The Internet and its Uses

5.1 The Internet and the WWW · 5.2 Digital Currency · 5.3 Cyber Security

5.1.1 The Internet and the World Wide Web (WWW)

The word **internet** comes from **INTERconnected NETwork**. It is a **worldwide collection of interconnected networks and devices**. It is a concept rather than something we can touch, and it relies on a **physical infrastructure** (cables, routers, satellites) that lets networks connect.

The **World Wide Web (WWW)** is only **a part of the internet**. It is a massive collection of **web pages** that users access with **web browser** software. The two are **not** the same thing.



The World Wide Web is only one part of the internet

Internet	World Wide Web (WWW)
A worldwide collection of **interconnected networks and devices**	A collection of **multimedia web pages** and other information on websites
Uses **TCP and IP** protocols	Uses **HTTP / HTTPS** protocols
Used for **email**, online chat (text, audio, video), file transfer and more	Web pages are written in **HTML**
Is the **infrastructure**	Uses the internet to get information from **web servers**
	Pages are located with **URLs** and viewed with a **web browser**

> **Exam Tip:** Email and online chat use the internet but are **not** part of the WWW. The WWW is **one service** that runs on the internet.

5.1.2 Uniform Resource Locators (URLs)

A **URL** is a **text address** used to access a website or web page. It is typed into the browser's **address bar**.

Format: `protocol://website address/path/file name`



The parts of a URL

Part	Meaning	Example
Protocol	The set of rules used. Usually **http** or **https**	`https`
Domain host	The type of host (usually www)	`www`
Domain name	The name of the website	`example`
Domain type	The kind of site: .com, .org, .net, .gov. Sometimes a **country code** follows (.uk, .de, .cy)	`.com`, `.co.uk`
Path	Where the web page is. If left out it is the **root directory** of the site	`/shop`
File name	The item on the web page	`index.html`

Together, the domain host, domain name, domain type and country code form the **website address**.

5.1.3 HTTP and HTTPS

Hypertext transfer protocol (HTTP) is a **set of rules that must be obeyed when transferring files** across the internet.

When security such as **SSL or TLS** is added, it becomes **HTTPS**. The **S stands for secure**. HTTPS sends and receives data in a more secure way (the data is **encrypted**). A **padlock** appears in the address bar.

	HTTP	HTTPS
Security	Data is sent as plain text, so it can be read if intercepted	Data is **encrypted** using SSL/TLS
Address bar	`http://...`, no padlock	`https://...` with a padlock
Use	General web pages with no private data	Online banking, shopping, logging in, any private data

5.1.4 Web Browsers

A **web browser** is software that lets a user **access and display web pages**. It **interprets (translates) the HTML** sent by a website and shows the result: text, images, video and audio.

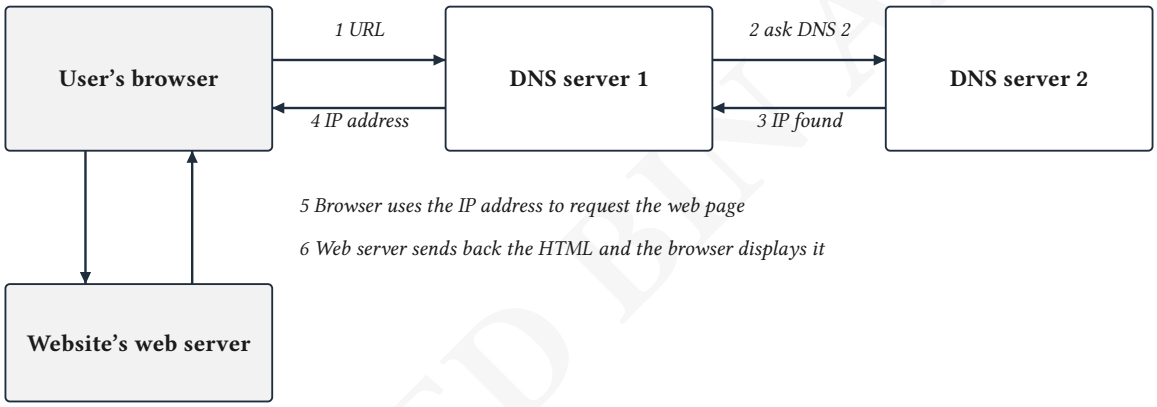
Common features of a browser:
- **Home page**: the page that opens when the browser starts.
- **Address bar**: where the URL is typed.
- **Bookmarks / favourites**: stores the user's favourite pages.

- **History**: a record of websites visited.
- **Back and forward** buttons to move through pages already opened.
- **Multiple tabs**: many pages open at the same time.
- **Hyperlinks**: click a link to move to another page (open in the same tab, or in a new tab with Ctrl + click).
- **Cookies**: stores user preferences and information (see 5.1.6).
- **Cache**: stores copies of pages so they load faster next time.
- Runs **JavaScript** on web pages.

5.1.5 Retrieval and Location of Web Pages

HTML (HyperText Markup Language) is the language used to **structure and display** web pages. A website is stored on a **web server**, and every web server has an **IP address**. The browser needs this IP address to get the page.

The **Domain Name Server (DNS)** (also called domain name system) **finds the IP address that matches a domain name**. DNS servers hold a **database of URLs and their IP addresses**. This means users do not need to remember IP addresses (for example 107.162.140.19).



How DNS is used to locate and retrieve a web page

Steps:

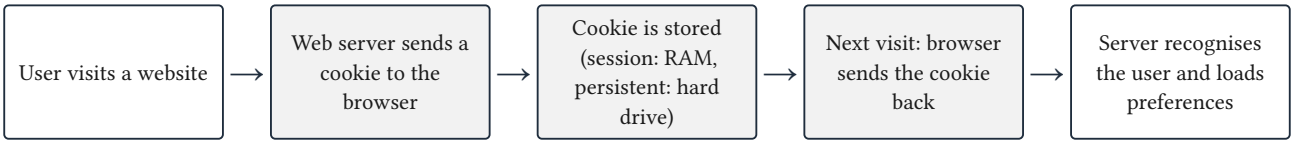
1. The user types the URL. The browser asks **DNS server 1** for the IP address.
2. If DNS server 1 cannot find it in its database or cache, it asks **DNS server 2**.
3. DNS server 2 finds the match and sends the IP address back. DNS server 1 **stores it in its cache** for next time.
4. The IP address is sent back to the user's computer.
5. The browser contacts the **web server** at that IP address and asks for the page.
6. The web server sends the **HTML files**. The **browser interprets the HTML** and displays the page.

> **Exam Tip:** Do not say "the DNS finds the web page". The DNS only gives the **IP address**. The **web server** sends the page.

5.1.6 Cookies

A **cookie** is a **small file or piece of code** sent by a web server to the user's browser and stored on the user's computer. A cookie is like a small **look-up table** of **(key, data)** pairs, for example (language, English) or (music, rock). Each time the user visits the site, the browser sends the cookie back and the site recognises the user.

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How a cookie is created and used

	Session cookie	Persistent (permanent) cookie
Stored in	**Temporary memory** (RAM)	The **hard drive**
Deleted when	The **browser is closed** or the session ends	The **expiry date** is reached or the **user deletes it**
Information	Does **not** collect information from the computer and does **not** identify the user	Remembers the user's **log-in details, preferences and browsing activity**
Typical use	**Virtual shopping basket** (keeps items while shopping)	Automatic log-in, saved preferences, targeted adverts

A cookie **with no expiry date** is always a **session cookie**.

Uses of cookies:
- Remember **passwords, email addresses and payment details** so the user need not type them again.
- Keep items in a **virtual shopping basket**.
- Store **user preferences** (language, currency, customised pages).
- **Track** browsing habits and history, and **target the user with adverts** that match their interests.
- Save progress in **online games and quizzes**.
- Used in **online financial transactions**.

> **Exam Tip:** Persistent cookies are called **tracking cookies** because they record browsing history and send it to advertisers. Always write the **type** of cookie when asked for differences (where it is stored and when it is deleted).

5.2.1 Digital Currency

Digital currency is money that **exists only in digital form**. It has **no physical form** (unlike **fiat currency** such as notes and coins) and is just **data on a computer system**. It is accepted as payment for goods and services and can be moved between accounts, for example with online banking, PayPal or a phone app such as Apple Pay.

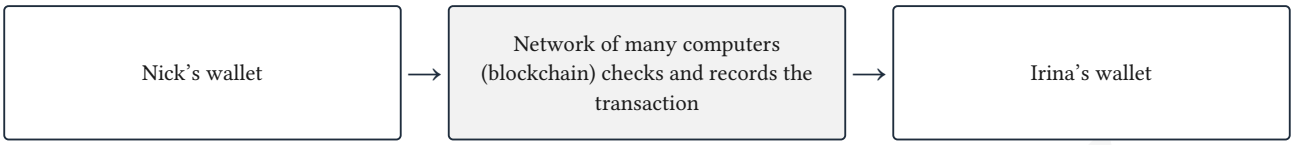
Normal digital currency is **centralised**: banks and governments regulate it and control all transactions and exchange rates. The problem with centralisation is **confidentiality and security**.

Cryptocurrency is a type of digital currency that fixes this by being **decentralised**. It uses **cryptography** to track and secure transactions and runs on a **blockchain** network.

Centralised digital currency (banks and central bank in control)



Cryptocurrency (decentralised, no bank in control)



Centralised digital currency and decentralised cryptocurrency

Digital currency (centralised)	Cryptocurrency (decentralised)
Controlled and **regulated by banks and governments**	**No state control**. Rules are set by the cryptocurrency community
A **central system** holds the records	Records are held on **many computers** in a blockchain network
Transactions are **not public**	Transactions are **publicly available** and can be tracked
Security and confidentiality issues with the central system	**More secure** because there is no central point to attack

5.2.2 Blockchain

A **blockchain** is a **decentralised database** in which all the transactions of the network members are stored. It is made of many **interconnected computers** but **no central server**. **Every computer holds a copy of every transaction**, so nothing can be changed without the agreement of the network.

Uses: cryptocurrency exchanges, smart contracts, research (e.g. pharmaceutical companies), politics (voting) and education.

What is in a block?

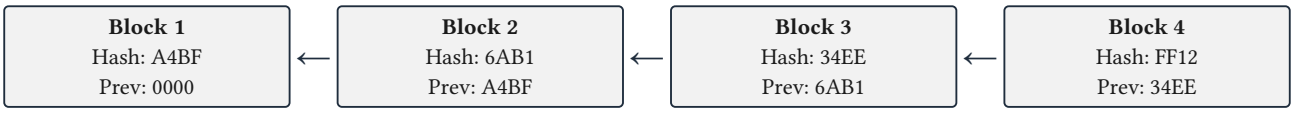
A new block is made for every new transaction. Each block stores:

- **Data**: for example sender, recipient and amount.
- **Hash value**: a **unique fingerprint** of the block, made by an algorithm (usually **SHA-256**). It includes a **timestamp** (the date and time the block was made).
- **Previous hash**: the hash of the block before it. This **links the blocks into a chain**.

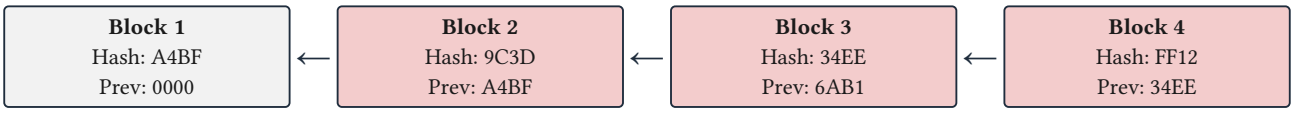
The first block is the **genesis block**. It has no previous block.

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A valid blockchain (each block stores the hash of the block before it)



Block 2 is tampered with: its hash changes, so the chain breaks (red = invalid)



Block 1 is the genesis block (it has no previous block). Block 3 still holds 6AB1, which no longer matches Block 2

Why is a blockchain hard to hack?

1. If a hacker **changes the data in one block**, its **hash changes**.
2. The next block now holds the **wrong previous hash**, so the **chain is broken** and that block and all later blocks become invalid. The tampering is spotted.
3. The hacker would have to **recalculate every later block** and do it on **every computer** in the network at the same time.
4. **Proof-of-work** slows this down. It takes about **ten minutes** of work to create the proof for each block before it can be added. Special users called **miners** do this work and receive a **commission** (reward) for each new block.
5. A new block is **sent to every computer** in the network and **checked** before it is added. A new user receives a **copy of the whole blockchain**.

> **Exam Tip:** For "how does blockchain protect against hackers", always include: **hash changes → chain breaks**, **copy on every computer**, and **proof-of-work slows down** any attempt to rebuild the chain.

5.3.1 Cyber Security Threats

Data can be lost, corrupted or stolen by accident or by **malicious acts**. The threats you must know are: **brute force attacks, data interception, DDoS attacks, hacking, malware, phishing, pharming and social engineering**.

Brute Force Attack

A hacker **systematically tries every possible combination** of letters, numbers and symbols until the password is found. It is simple and not clever, but it works on weak passwords.

To make it faster, the attacker first:
1. tries the **most common passwords** (for example 123456, password, qwerty, 111111, abc123)
2. then uses a **word list** (a text file of many words), which is faster than total trial and error.

Prevention: use **long, strong passwords** with a mix of characters, **limit the number of log-in attempts** (account locks after three failures), and use two-step verification.

Data Interception

Stealing data by tapping into a wired or wireless communication link, to find private or confidential information.

- **Packet sniffing:** a program **examines the data packets** sent over a network and the intercepted data is sent back to the hacker. Common on wired networks.

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- **Wardriving (access point mapping):** using a **laptop or smartphone, an antenna, GPS and software** near a building to pick up **Wi-Fi signals** and read the data, often without the user knowing.

Prevention: encryption (the data is meaningless without the decryption key), a **firewall**, a wireless encryption protocol (WEP, or the stronger WPA), a **strong router password**, and **not using public Wi-Fi** for private data.

> Encryption does **not stop** data being intercepted. It only makes the data **unreadable** to the hacker.

Distributed Denial of Service (DDoS) Attack

A **denial of service (DoS)** attack tries to **stop users from reaching a website, email or online service** by flooding it with useless requests. A web server can only handle a limited number of requests, so when thousands arrive it cannot serve **genuine** requests.

In a **DDoS** attack the requests come from **many different computers** (often infected with malware), which makes the attack **hard to block**.

Signs of an attack: slow network performance, unable to open certain websites, a huge amount of spam arriving in the mailbox.

Prevention: an **up-to-date malware checker**, a **firewall** to restrict traffic, **email filters** to remove spam, and a **proxy server**.

Hacking

Gaining illegal access to a computer system without the owner's permission. Data can be **deleted, changed, corrupted or passed on**, and it can lead to **identity theft**.

Prevention: firewalls, **strong passwords that are changed often**, user names, **anti-hacking and intrusion-detection software**. Encryption does not stop hacking, but it makes the stolen data meaningless.

Ethical hacking is when a company **pays and authorises** a hacker to test how strong its security is. This is legal. Hacking without permission (**malicious hacking**) is **always illegal**.

Malware

Malware (malicious software) is installed on a computer to **delete, corrupt, steal or manipulate data**.

Malware	What it does	Key points
Virus	A program or code that **replicates itself** to delete or corrupt files or make the computer malfunction	Needs an **active host** (a program or infected OS) and must be **run** by a trigger. Often arrives as **email attachments**, infected websites or software
Worm	A **stand-alone** program that **self-replicates** and spreads to other computers and **whole networks**	Needs **no host** and **no user action**. Uses **weak network security**. More dangerous than a virus
Trojan horse	Looks like **legitimate software** but contains **malicious code**	The **user must run it**, so it gets past firewalls. Often delivers spyware or ransomware. Example: **fake anti-virus** pop-up
Spyware	**Monitors the user's activity** and sends the information to the criminal (passwords, card numbers)	Includes **key logging software**, which records key presses. Detected by **anti-spyware**

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Adware	Floods the user with **unwanted adverts** (pop-ups, redirected searches, toolbar changes)	Not always harmful, but it can **hijack the browser** and show security weaknesses. Hard to remove
Ransomware	**Encrypts the user's data** and demands a **ransom** for the decryption key	Spread by **Trojan horses** or social engineering. Damage is almost impossible to reverse, so **regular back-ups** are the best defence

> **Exam Tip: Virus vs worm:** a virus needs a **host program and a user to run it**. A worm **spreads by itself** across a network with **no user action**. Use these words in your answer.

Defence against all malware: up-to-date **anti-virus / anti-malware**, a **firewall**, do not open **attachments or links from unknown senders**, do not install **non-original software**, and keep **regular back-ups**.

Phishing

The criminal sends **legitimate-looking emails** pretending to be from a bank or service provider. The email contains **links or attachments** that lead to a **fake website**, or it tricks the user into **sending personal data** (bank or card details). The user **has to act** (click or reply) for the attack to work.

Spear phishing is aimed at **specific people or companies**, usually to reach financial information or for industrial spying. Ordinary phishing is not aimed at anyone in particular.

Prevention:
- **Security awareness training** for staff.
- **Never click links** in an email unless you are sure they are safe. Watch for greetings such as "Dear Customer".
- Use **anti-phishing toolbars** and an **up-to-date browser**.
- Look for **https and the padlock**.
- Run a **firewall** and **block pop-ups**. Do not click "Cancel" on a suspicious pop-up; close it using the small X.
- Check online accounts regularly and **change passwords**.

Pharming

Malicious code installed on the user's computer or on a website that **redirects the browser to a fake website** without the user knowing. The **user does not have to do anything** for it to start. The fake site collects bank details and passwords, leading to fraud and identity theft.

DNS cache poisoning: the attacker **changes the IP address stored on a DNS server** to the IP address of the fake site. When the user types the real URL, the DNS sends the **wrong IP address** and the browser connects to the fake website.

Prevention: anti-virus software (detects unauthorised changes), a **firewall**, a **modern browser** that warns of pharming, **checking the spelling** of the web address, and **https and the padlock**.

	Phishing	Pharming
How it starts	A fake **email** with a link or attachment	**Malicious code** on the computer, or **DNS cache poisoning**
User action needed?	**Yes**: the user must click or reply	**No**: the redirect happens automatically

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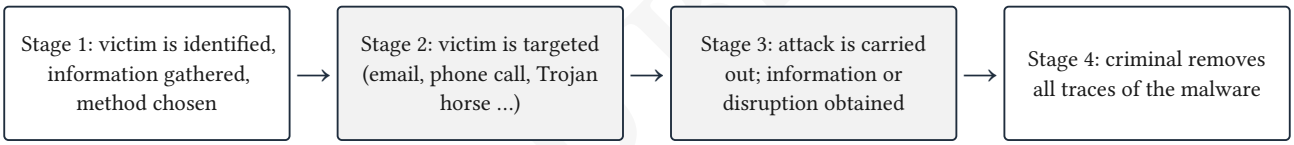
Result	User is sent to a **fake website** or gives data away	User is sent to a **fake website** even though they typed the **correct URL**

Social Engineering

The criminal **manipulates people** into **breaking normal security procedures**. There is **no hacking**: the user willingly gives access or runs the malware. It works by exploiting **human emotions**.

Method	How it works	Emotion used
Instant messaging	A **malicious link** is sent in a message (for example an "important software upgrade")	Curiosity
Scareware	A pop-up claims the computer has a virus and tells the user to download a **fake anti-virus**	Fear
Emails / phishing	A genuine-looking email contains a link to a **fake website**	Trust
Baiting	A **malware-infected memory stick** is left where it will be found. The finder plugs it in to see who owns it	Curiosity
Phone calls	A fake "IT professional" says the device is compromised and tells the user to install software that gives the criminal control	Fear

Emotions exploited: fear (panic, no time to think), **curiosity** (a prize or a found memory stick) and **empathy / trust** (belief that well-known companies are safe).



Stages in a typical social engineering attack

5.3.2 Keeping Data Safe from Security Threats

Access Levels

User accounts give **different levels of access** to different people, usually with a **user name and password**. For example, in a hospital a consultant can see patient records but a cleaner cannot. In a database, access levels decide who can **read, write and delete** data, and **views** can show users only some of the data.

Social networks use **four levels**: **public**, **friends**, **custom** and **data owner**.

Anti-malware

Anti-virus was covered in Chapter 4 (checks files against a database of known viruses, uses heuristics, quarantines infected files).

Anti-spyware detects and removes spyware. It works using **rules** (typical spyware behaviour) or **known file structures**. Its features:

- detects and removes spyware already installed
- **prevents spyware being downloaded**
- **encrypts keyboard strokes** to defeat key loggers
- **blocks access to the webcam and microphone**
- warns the user if personal information has been stolen.

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Today anti-spyware is often part of a **bundle** with anti-virus and a personal firewall.

Authentication

Authentication is the ability of a user to **prove who they are**. There are three factors:

- **something you know** (password, PIN)
- **something you have** (mobile phone, tablet, card)
- **something unique to you** (biometrics).

Passwords and user names

A password must be **hard to guess**. Passwords are **weak** (easy to guess, e.g. a pet's name or `GREEN`) or **strong**.

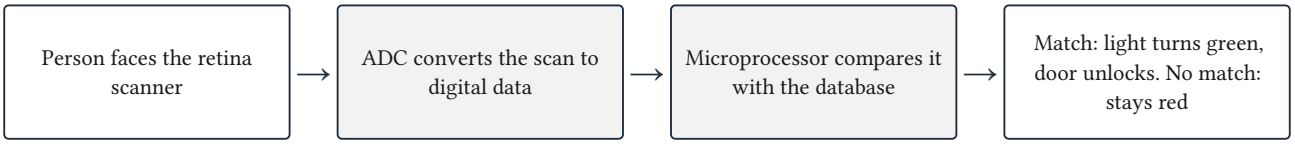
- A **strong password** is **long** and contains **capital letters, numbers and other keyboard symbols**, for example `Sy12@#TT90kj=0`.
- Do **not** use **personal information** (birthdays, pet names) or common words.
- **Change passwords regularly**, and run **anti-spyware** so passwords are not sent to a criminal.
- Passwords show as ******** on screen so nobody can read them.
- Users get only a **limited number of attempts** (usually three) before the account is **locked**.
- A forgotten password is reset using a **link sent by email**.
- **Never let a device save passwords.**

Biometrics

Biometrics uses **unique human characteristics** to identify a user: **fingerprints, retina scans, face recognition and voice recognition**. Fingerprints are compared with stored images by matching the pattern of **ridges and valleys** (accuracy about 1 in 5 000). A **retina scan** uses **infrared light** to map the **blood vessel pattern** at the back of the eye (accuracy about 1 in 10 million).

Benefits of biometrics in general: they are **unique**, they **cannot be lost or forgotten**, they are **hard to copy**, and nobody can sign in for someone else.

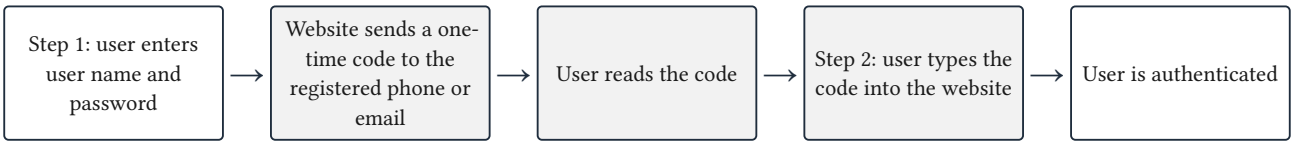
Technique	Benefits	Drawbacks
Fingerprint	• Very developed technique • Easy to use • Small storage needed	• Can be **intrusive** (linked with crime) • Dirty or **damaged skin** can cause errors • **Expensive** to install
Retina scan	• **Very high accuracy** • No known way to copy a retina	• **Intrusive** and unpleasant (sit still 10 to 15 seconds) • **Slow** to verify • **Very expensive**
Face recognition	• **Non-intrusive** • Relatively cheap	• Affected by **lighting, hair, age and glasses**
Voice recognition	• **Non-intrusive** • Fast (under 5 seconds) • Relatively cheap	• Voice can be **recorded and replayed** • **Low accuracy** • A **cold** changes the voice



A biometric door security system

Two-step verification

The user must pass **two methods of authentication**. It is used mainly when **paying online with a card**. The phone is **something you have** and the password is **something you know**.



Two-step verification

Automatic Software Updates

Software on computers and phones is kept **up to date automatically** (often overnight). Updates contain **patches** that fix **security weaknesses** (protecting against malware) and **bugs**, and may add features.

Downside: an update can sometimes **disrupt the device**. The user then has to wait for another patch or use **system restore** to go back to an earlier state.

Checking Emails and URL Links

Before opening an email or clicking a link, check:

1. **The sender's email address.** The part after the `@` should match the real company. A company will not use a free address such as a Gmail address.
2. **Spelling and grammar.** Professional organisations do not send emails with spelling mistakes.
3. **Tone.** A message that **rushes or threatens** you ("act immediately") is suspicious. Greetings like "Dear Customer" are a warning sign.
4. **Misspelt domain names.** This is **typo squatting**: names close to the real one are used to fool you (for example `amozon.com`).
5. **Suspicious links.** Hover over a link to see its real destination. It should match the company. Secure sites should use **https**.

Firewalls

A **firewall** can be **software or hardware**. It sits **between the user's computer (or network) and an external network** such as the internet and **filters all traffic in and out**. It is the **first line of defence** against hacking, malware, phishing and pharming.



Data that fails the rules is blocked and a warning is given

A firewall sits between the computer and the internet

Tasks of a firewall:

- **Examines** all incoming and outgoing traffic.
- **Checks** whether the data meets a **set of rules (criteria)**.
- **Blocks** data that fails the rules and **warns** the user or network manager.
- **Logs** traffic so it can be checked later.
- **Blocks access to undesirable sites** (keeps a list of bad IP addresses).
- Warns the user if a program tries to **access the internet**, and lets the user allow or deny it.

Limits of a firewall (it cannot stop):

- users on the internal network using their **own devices** (modems, smartphones) to bypass it
- **carelessness or misconduct** by employees (for example sharing passwords)

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- a user on a stand-alone computer **turning the firewall off**.

Proxy Servers

A **proxy server** acts as an **intermediary** between the user and the web server.



The proxy filters traffic, hides IP addresses and stores (caches) pages

A proxy server sits between the user and the web server

Features:

- **Filters** internet traffic. It can **block** certain websites or certain IP addresses.
- Valid traffic is passed on to the web server. **Invalid traffic is refused.**
- **Hides the user's IP address**, which improves security.
- **Prevents direct access** to the web server, so attacks (hacking, DoS) **hit the proxy first**.
- Uses a **cache**: it stores copies of web pages, so the next request for the same page is served **faster** from the proxy.
- Can also act as a **firewall**.

Privacy Settings

Privacy settings are controls on browsers, social networks and websites that **limit who can see a user's data and what is collected**. Examples:

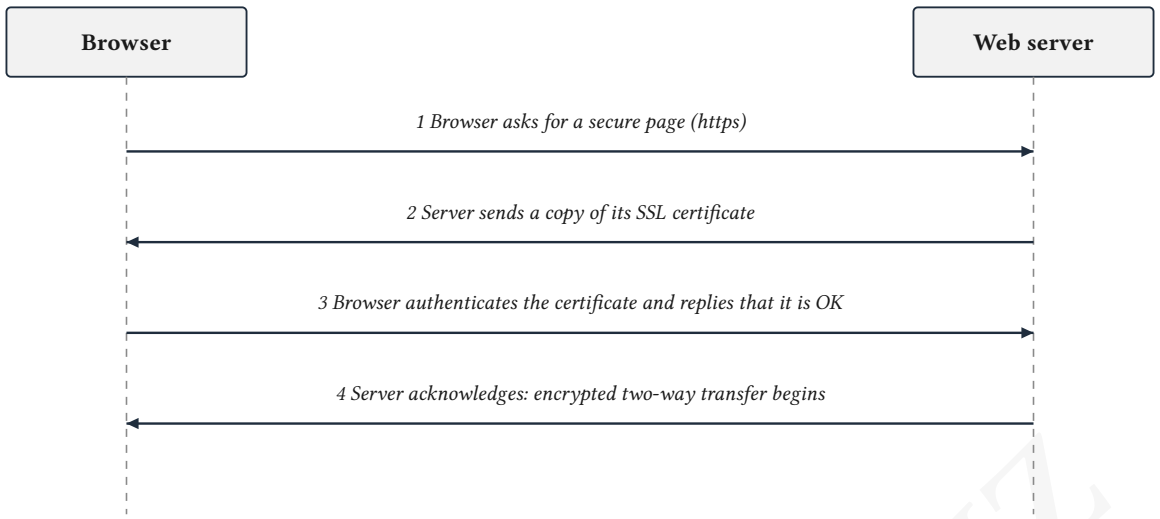
- a **"do not track"** setting that stops websites collecting browsing data
- **safer browsing** alerts for dangerous websites
- browser options such as **storing history and cookies**
- **advertising opt-outs**
- **switching off location sharing** in apps
- checking whether **payment details are saved** on websites (saved details can be stolen).

Secure Sockets Layer (SSL) and TLS

SSL is a **protocol (set of rules)** that lets data be sent and received **securely** over the internet. It **encrypts** the data so only the user's computer and the web server can understand it. The user knows SSL is in use when the address starts with **https** and a **padlock** appears.

An **SSL certificate** is a **digital certificate** that **authenticates a website**. If the certificate can be authenticated, communication is secure.

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How SSL sets up a secure connection

TLS (Transport Layer Security) is the **newer and more secure version of SSL**. It gives **encryption, authentication and data integrity** and stops third parties eavesdropping.

SSL / TLS is used for: online banking and financial transactions, online shopping, email, cloud storage, instant messaging, VoIP, social networking, and sending software to a restricted list of users.

> **Exam Tip:** SSL question: **browser asks for a secure page → server sends certificate → browser authenticates it → server acknowledges → encrypted transfer begins.** Always mention the **SSL certificate** and **encryption**.

Summary: Threat and Prevention

Threat	Main ways to prevent it
Brute force	Strong, long passwords; limit log-in attempts; two-step verification
Data interception	Encryption; firewall; strong router password; avoid public Wi-Fi
DDoS	Firewall; proxy server; email filters; up-to-date malware checker
Hacking	Firewall; strong passwords changed often; intrusion-detection software
Malware	Anti-virus and anti-spyware; firewall; do not open unknown attachments; back-ups; updates
Phishing	Do not click unknown links; training; anti-phishing toolbar; check https and spelling
Pharming	Anti-virus; firewall; check the web address; https and the padlock
Social engineering	Staff training; be suspicious of urgent or unexpected requests; do not plug in unknown memory sticks

Model Answers

Give the difference between the internet and the WWW.

- The internet is a worldwide collection of interconnected networks and devices.
- The WWW is a collection of web pages that are accessed using a web browser over the internet.

Describe how a web page is retrieved using a URL.

- The browser sends the URL to a DNS server, which finds the matching IP address.
- The browser uses the IP address to contact the web server, which sends the HTML files.
- The browser interprets the HTML and displays the page.

Give two differences between a session cookie and a persistent cookie.
- A session cookie is stored in temporary memory, but a persistent cookie is stored on the hard drive.
- A session cookie is deleted when the browser is closed. A persistent cookie stays until its expiry date or until the user deletes it.

Explain how a blockchain protects against hacking.
- Each block contains a hash and the hash of the previous block.
- If a block is changed its hash changes, so the next block no longer matches and the chain is broken.
- Every computer holds a copy and proof-of-work makes creating new blocks slow, so a hacker cannot easily rebuild the chain.

Describe how a firewall helps protect a computer.
- It examines all traffic entering and leaving the computer or network.
- It checks the traffic against a set of rules and blocks data that fails them.
- It can block access to certain websites or IP addresses, and keeps a log of traffic.
- It warns the user when a program tries to access the internet.

What is meant by phishing? How is it different from pharming?
- Phishing is a fake email that tricks the user into clicking a link or giving personal data. The user must take an action.
- Pharming is malicious code or DNS cache poisoning that redirects the user to a fake website without any action from the user.

Describe how two-step verification works.
- The user enters their user name and password.
- A one-time code is sent to the registered phone or email.
- The user enters the code. If it is correct, the user is authenticated.

Describe how SSL provides a secure connection.
- The browser requests a secure page and the web server sends its SSL certificate.
- The browser authenticates the certificate and tells the server.
- The server acknowledges and all data is then encrypted in both directions, so it cannot be understood if intercepted.

Describe how a proxy server improves security.
- It sits between the user and the web server and filters traffic, blocking invalid requests.
- It hides the user's IP address and stops attacks reaching the web server directly.
- It stores a cache of web pages so they load faster.

Common Exam Questions

- Give the differences between the internet and the WWW.
- Identify the protocol, domain name, domain type and file name in a URL.
- State the purpose of HTTP and HTTPS, and what the "S" means.
- Describe the purpose and features of a web browser.
- Describe how DNS is used to find and retrieve a web page.
- Describe session cookies and persistent cookies, and give uses of cookies.
- Compare digital currency and cryptocurrency.
- Explain how a blockchain works and how it protects data.

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- Describe brute force, data interception, DDoS and hacking, and how to prevent each.
- Describe viruses, worms, Trojan horses, spyware, adware and ransomware.
- Compare phishing and pharming, and explain DNS cache poisoning.
- Describe social engineering, the emotions it uses and its four stages.
- Describe access levels, authentication, passwords, biometrics and two-step verification.
- Describe how a firewall and a proxy server work.
- Describe how SSL / TLS and an SSL certificate keep data safe.
- Describe signs of a suspicious email or URL.

Quick Revision

- **Internet** = worldwide network of networks (TCP/IP). **WWW** = web pages accessed with a browser (HTTP/HTTPS, HTML, URLs).
- **URL** = protocol, domain host, domain name, domain type (country code), path, file name.
- **HTTPS** = HTTP with SSL/TLS encryption and a padlock.
- **DNS** finds the IP address of a domain name. The **web server** sends the HTML. The **browser** displays it.
- **Session cookie:** temporary memory, deleted when the browser closes. **Persistent cookie:** hard drive, kept until expiry or deletion.
- **Digital currency** is electronic and centralised. **Cryptocurrency** is decentralised and uses a **blockchain**.
- **Block** = data + hash + previous hash + timestamp. Changing a block breaks the chain. **Proof-of-work** and **miners** slow down attacks.
- **Brute force:** trial of all combinations. **Interception:** packet sniffing and wardriving, defeated by encryption. **DDoS:** flood of requests from many computers.
- **Virus** needs a host and a user. **Worm** spreads alone. **Trojan** is disguised software. **Spyware** records activity. **Adware** shows adverts. **Ransomware** encrypts data for money.
- **Phishing** = fake email, user acts. **Pharming** = redirect to fake site (DNS cache poisoning), no user action.
- **Social engineering** uses fear, curiosity and trust: scareware, baiting, phone calls, fake emails and messages.
- **Authentication:** something you know, something you have, something unique to you (biometrics). **Two-step verification** = password + one-time code.
- **Firewall** filters traffic against rules. **Proxy server** hides IP addresses, filters and caches pages.
- **SSL/TLS** uses a digital certificate and encryption. Look for **https** and the padlock.
- **Updates (patches)** fix security holes. **Back-ups** defeat ransomware.